

Lecture-3

We saw in the last lecture, how to solve a system of two equations and two variables. In this lecture we shall introduce the concept of a matrix, which can be used effectively to solve a system of m -equations and n -variables

Matrix

A matrix is an array of numbers written in the form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1j} & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2j} & a_{2n} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{ij} & a_{in} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mj} & a_{mn} \end{pmatrix}$$

j^{th} Column

i^{th} row

Entry from i^{th} row and j^{th} column

The above matrix is called an $m \times n$ matrix: m - rows
 n - columns

Matrix addition

Any two $m \times n$ matrices can be added as follows

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \dots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \dots & a_{2n}+b_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \dots & a_{mn}+b_{mn} \end{pmatrix}$$

Matrix Multiplication

An $m \times n$ matrix and an $n \times k$ matrix can be multiplied.

Before we define multiplication, here is a notation

If $a_1, a_2, \dots, a_n$ are numbers then for convenience we shall write $\sum_{i=1}^n a_i$ as a short hand for $a_1 + a_2 + \dots + a_n$

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1k} \\ b_{21} & b_{22} & \dots & b_{2k} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nk} \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n a_{1j} b_{j1} & \sum_{j=1}^n a_{1j} b_{j2} & \dots & \sum_{j=1}^n a_{1j} b_{jk} \\ \vdots & \vdots & & \vdots \\ \sum_{j=1}^n a_{mj} b_{j1} & \sum_{j=1}^n a_{mj} b_{j2} & \dots & \sum_{j=1}^n a_{mj} b_{jk} \end{pmatrix}$$

When an $m \times n$ matrix is multiplied by an $n \times k$ matrix, we get an $m \times k$ matrix.

Examples

$$\begin{pmatrix} 4 & 3 & 2 \\ 1 & 7 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 0 \\ -1 & 2 & -4 \end{pmatrix} = \begin{pmatrix} 5 & 7 & 2 \\ 0 & 9 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 3 & 2 \\ 1 & 7 & 4 \end{pmatrix} \begin{pmatrix} 1 & 4 & 0 \\ -1 & 2 & -4 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 4-3+2 & 16+6+4 & -12+6 \\ 1-7+4 & 4+14+8 & -28+12 \end{pmatrix}$$

2×3
matrix

3×3
matrix

$$= \begin{pmatrix} 3 & 26 & -6 \\ -2 & 26 & -16 \end{pmatrix}$$

2x3 matrix.

⊛ An $n \times n$ matrix is called a Square matrix.

Representation of a System of Equation.

A System of equation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

Can be represented as a "matrix equation" as follows

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

Identity matrix :

An $n \times n$ matrix of the form

$$I_n := \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

$$a_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Inverse of an $n \times n$ matrix:

A square matrix A is said to have an inverse if there is another square matrix B such that

$$AB = I_n = BA.$$

Examples.

$$\textcircled{1} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{Also } \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus the inverse of $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ is $\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$ and

Vice Versa.

$\textcircled{2}$ The matrix $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ has no inverses!

Note that $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a_{11} + a_{12} & 0 \\ a_{21} + a_{22} & 0 \end{pmatrix}$, which will

never become the identity matrix for any choice of $a_{11}, a_{12}, a_{21}, a_{24}$.

Notation: ① We shall write

$$AX = B$$

is a system of equations, where it is understood that A is an $m \times n$ matrix, X is an $n \times 1$ matrix and B is an $m \times 1$ matrix.

② Entries of a matrix A are denoted by a_{ij} . So, we sometime write (a_{ij}) to denote the matrix

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

③ $I_n X = B$, where $X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, $B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$

Then

$$\begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \dots & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

And therefore, we have $x_1 = b_1, x_2 = b_2, \dots, x_n = b_n$.

Existence of inverse and Unique Solution

For the system $AX = B$, where A is an $n \times n$ matrix, Suppose that A has an inverse.

Say $AC = I_n = CA$. Then, we have

$$CAX = CB.$$

That is, $I_n X = CB$. This implies

$X = CB$, which is a unique solution.

Example

Consider the System

$$2x + 5y = 1$$

$$x + 3y = 0$$

Then, its matrix form is given by

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Since $\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$ is the inverse of $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$,

$$\text{we have } \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

Thus, $x = 3, y = -1$. Of course, the solution is unique

* If inverse exists then it is Unique.

If $AB = I_n = BA$ and $AC = I_n = CA$ then $B = C$.

Solution

$$\begin{aligned} AB = I_n &\Rightarrow (AB)C = I_n C = C \\ &\Rightarrow (BA)C = C \quad \left[\text{Since } AB = BA = I_n \right] \\ &\Rightarrow B(AC) = C \\ &\Rightarrow B I_n = C \\ &\Rightarrow B = C. \end{aligned}$$

Unique — "One and only one"

Some properties of matrix multiplication.

Let $X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$. For a number α , let αX be the matrix $\begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}$.

Let A be an $n \times n$ matrix, X and Y be $n \times 1$ matrix and α be a number. Then,

$$\textcircled{1} \quad A(X + Y) = AX + AY.$$

$$\textcircled{2} \quad A(\alpha X) = \alpha AX$$

Proof:

Note that

$$AX = \begin{pmatrix} \sum_{j=1}^n a_{1j} x_j \\ \vdots \\ \sum_{j=1}^n a_{nj} x_j \end{pmatrix}$$

and that

$$AY = \begin{pmatrix} \sum_{j=1}^n a_{1j} y_j \\ \vdots \\ \sum_{j=1}^n a_{nj} y_j \end{pmatrix}$$

$$\begin{aligned} \text{Then, } AX + AY &= \begin{pmatrix} \sum_{j=1}^n a_{1j} x_j + \sum_{j=1}^n a_{1j} y_j \\ \vdots \\ \sum_{j=1}^n a_{nj} x_j + \sum_{j=1}^n a_{nj} y_j \end{pmatrix} \\ &= \begin{pmatrix} \sum_{j=1}^n a_{1j} (x_j + y_j) \\ \vdots \\ \sum_{j=1}^n a_{nj} (x_j + y_j) \end{pmatrix} \\ &= A(X + Y) \end{aligned}$$

For ②, we observe that

$$A(\lambda X) = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n \lambda a_{1j} x_j \\ \vdots \\ \sum_{j=1}^n \lambda a_{nj} x_j \end{pmatrix}$$

$$= \alpha \begin{pmatrix} \sum_{j=1}^n a_{1j} x_j \\ \vdots \\ \sum_{j=1}^n a_{nj} x_j \end{pmatrix} = \alpha A X.$$